

DecodeLabs | Batch 2026
Data Analytics — Project 3

SQL Data Analysis

Database: decodelabs | Table: p3 | Tool: SQL Server 16

1. Project Overview

Project 3 focuses on querying a relational database using SQL to extract business insights. The cleaned dataset from Project 1 was imported into SQL Server and analysed using SELECT, WHERE, GROUP BY, HAVING, ORDER BY, and aggregate functions (COUNT, SUM, AVG, MAX, MIN).

Database	deodelabs
Table	p3
Total rows	1,200
Total columns	14
Tool	SQL Server Management Studio 21
Queries written	15

2. Revenue Overview

Basic aggregation across all 1,200 orders:

total_orders	total_revenue	avg_order	highest	lowest
1200	1,264,761.96	1053.97	3456.40	11.39

Total revenue is **\$1,264,761.96**. The average order is \$1,053.97. The gap between highest (\$3,456) and lowest (\$11.39) shows a wide spread of order values.

3. Product Analysis

Orders and revenue by product, ordered by revenue:

Product	orders	revenue	avg_value
Chair	178	195,620.11	1098.99
Printer	181	195,612.61	1080.73
Laptop	173	192,126.56	1110.56
Tablet	179	186,568.95	1042.28
Monitor	163	175,651.41	1077.62
Desk	170	167,459.93	985.06
Phone	156	151,722.39	972.58

Chair and **Printer** lead in total revenue (~\$195K each). **Laptop** has the highest average order value at \$1,110.56. **Phone** has the fewest orders (156) and lowest revenue (\$151K).

4. Payment Method Analysis

How customers pay:

PaymentMethod	total	pct
Online	258	21.5%
Cash	246	20.5%
Credit Card	234	19.5%
Debit Card	232	19.3%
Gift Card	230	19.2%

All five payment methods are nearly equal (19–21%). No single method dominates, so all channels must be kept friction-free.

5. Order Status Analysis

Status breakdown across all orders:

OrderStatus	total	pct
Cancelled	250	20.8%
Returned	247	20.6%
Pending	237	19.8%
Shipped	235	19.6%
Delivered	231	19.3%

Only 19.3% of orders are Delivered. Cancelled + Returned = 41.4% — a critical operational issue. Less than 1 in 5 orders reaches the customer successfully.

6. Cancelled Orders — Lost Revenue

cancelled	lost_revenue	avg_value
250	276,396.21	1105.58

\$276,396 lost to cancellations. The average cancelled order (\$1,105) is higher than the overall average (\$1,053) — cancellations disproportionately affect higher-value orders.

7. Referral Source Analysis

Where customers come from:

ReferralSource	orders	revenue
Instagram	259	275,285.45
Email	250	261,808.55
Google	241	250,441.48
Facebook	228	250,410.90
Referral	222	226,815.58

Instagram leads with 259 orders and \$275K revenue. Word-of-mouth Referral has the lowest numbers but is worth growing — referred customers tend to be high-intent.

8. Monthly Revenue Trend (first 8 months)

month	orders	revenue
2023-01	47	56,685.75
2023-02	37	40,117.66
2023-03	43	48,609.37
2023-04	31	27,751.71
2023-05	49	63,836.84
2023-06	45	49,500.19
2023-07	44	42,820.66
2023-08	51	54,352.14

May 2023 peaked at \$63,836 (49 orders). April 2023 was the lowest at \$27,751 (31 orders). Full dataset covers 30 months from Jan 2023 to Jun 2025.

9. Top 8 Highest Value Orders

OrderID	Product	TotalPrice	OrderStatus
ORD200789	Tablet	3456.40	Delivered
ORD201122	Monitor	3390.95	Returned
ORD200632	Laptop	3390.80	Delivered
ORD200469	Chair	3384.90	Cancelled
ORD200328	Tablet	3370.20	Cancelled
ORD200107	Printer	3353.75	Shipped
ORD200326	Laptop	3352.40	Returned
ORD201065	Printer	3334.00	Delivered

Of the top 8 orders by value, only 3 were Delivered. 2 were Cancelled, 2 Returned. High-value orders have a disproportionately high failure rate.

10. Coupon Code Usage

CouponCode	used	revenue
FREESHIP	313	335,036.99
NO_COUPON	309	322,401.41

WINTER15	292	302,483.54
SAVE10	286	304,840.02

FREESHIP is the most used coupon (313 times) and generates the highest revenue. Orders with no coupon still account for \$322K — a segment that buys without incentives.

11. Units Sold per Product

Product	total_units	avg_qty
Chair	562	3.16
Printer	542	2.99
Laptop	535	3.09
Desk	508	2.99
Tablet	497	2.78
Monitor	480	2.94
Phone	411	2.63

Chair leads in total units (562, avg 3.16 per order). **Phone** has the lowest units (411) and lowest average quantity (2.63).

12. Key Findings & Recommendations

Cancellation + Return rate = 41.4%

Over 4 in 10 orders never complete. Biggest operational problem in the dataset. Needs root cause investigation — payment failures, stock, or UX.

Lost revenue from cancellations = \$276,396

Cutting the cancel rate by 50% would recover ~\$138K. Highest-impact fix available.

Instagram is the top acquisition channel

259 orders, \$275K revenue. Clearest ROI for marketing budget.

FREESHIP drives the most orders

313 uses and highest revenue. Free shipping removes a key purchase barrier.

Laptop has the highest avg order value (\$1,110)

Worth prioritising in upsell and bundle strategies despite lower order count.

Only 19.3% of orders are Delivered

Less than 1 in 5 orders reaches the customer. Fixing fulfilment is urgent.